

Cognitive Interviewing: A Tool for Questionnaire Development and Evaluation

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Session Outline

- Complexities of survey measurement
- What is cognitive interviewing and how can it help?
- Practical considerations: “how to do it”
- Some examples of design and analysis
- Maximizing the usefulness of the methodology

Survey data: essential, and rooted in principles of basic human communication

- Provide information about health, education, employment, economic behavior, political attitudes... essential for policy planning, decision making, and smooth functioning of democracy
- Are in many cases the only plausible or practical source of this greatly valued information
- Are derived from **question and answer**, a common form of human communication and information exchange, which we successful use constantly in everyday life
- Still, they are subject to numerous threats to *data validity*

Potential complications with questions: inherent complexity of language

▶ Conceptual problems

- ▶ Ambiguous language– could answer differently depending upon interpretation
- ▶ Too complex to fully absorb
- ▶ Based on incorrect assumptions
- ▶ Unknown terms

▶ Limitation of human knowledge, and memory

- ▶ Respondent memories subject to inaccuracy and distortion
- ▶ They might have never known what we want them to report

▶ Unknown and unintended influences on interpretations

- ▶ Loaded questions that suggest a “correct” answer
- ▶ Prior questions affect your interpretation of latter ones

Potential complications (2)

- ▶ Poorly chosen response categories
 - ▶ Yes/No questions that depend on various situations
 - ▶ No response category that matches respondent experience
- ▶ Sensitivity: might not want to accurately disclose information
- ▶ Motivation: responding accurately isn't worth the trouble
- ▶ Mode of delivery can affect understanding of the response task and ability to complete it
- ▶ Multiple languages, variation in regional and cultural terminology and understanding

How much do these things matter?

- ▶ Two surveys assess how many people have difficulty reading typical print:
- ▶ National Health Interview Survey (NHIS):
 - ▶ Wearing your (glasses/contact lenses) if that is how you see best . . . can you see well enough to read newspaper print?
- ▶ Survey of Income and Program Participation (SIPP)
 - ▶ Do you have difficulty seeing the words and letters in ordinary newspaper print even when wearing glasses or contact lenses, if you usually wear them?
- ▶ Resulting estimates:
 - ▶ NHIS: 4.2 million
 - ▶ SIPP: 9.7 million

Why such different results?

The questions look similar– but....

- ▶ Subtle wording differences matter (“unable to see well enough to read” vs “difficulty seeing words and letters”)
- ▶ In a follow-up study, some reported that they *could see well enough to read*, but still had *difficulty seeing words and letters*.
- ▶ Note that one question is framed positively and another negatively
 - ▶ NHIS: “yes” means *can read*
 - ▶ SIPP: “yes” means *have difficulty*
- ▶ The questions asked just prior to these are different on the two surveys, potentially leading to context effects (prior vision questions => higher self-evaluation)

Problems answering questions might not be obvious

- Respondents may simply provide an answer without revealing any misunderstandings or errors
 - In the past year, have you been bothered by pain in the abdomen?
 - Do you currently smoke cigarettes: daily, occasionally, or not at all?
- Methods of survey interviewing suppress expressions of confusion or uncertainty, pushing respondents toward codeable answers:
 - Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?

Different approaches can illuminate the *extent* or the *reasons* for measurement problems

- ▶ **Quantitative methods** (e.g., experiments): identify statistical differences and consequences of various design alternatives
- ▶ **Qualitative methods** (e.g., cognitive interviewing): identify problems and why questions and instruments have error
- ▶ Both are useful, but the qualitative options are (usually) cheaper, faster, and better evaluation tools for finding *conceptual weaknesses that may not be obvious*

A practical, efficient, and valid question evaluation tool should:

- Illuminate issues that could threaten the quality of survey responses, which may not be readily visible
- Realistically reflect real survey experience
 - Involve human respondents, similar to the ones we expect will answer the actual survey
 - Approximate the conditions of the actual survey
- Be relatively easy to carry out
 - Require reasonable resources
 - Can be done in a timely manner

Cognitive Interviewing accomplishes this.

In practice:

- 1) We administer draft survey questions to individual respondents and observe how they answer...
- 2) In addition to their response, we get them to provide additional verbal information about their responses...
- 3) And use this material to:
 - Evaluate the quality of responses
 - Or help to determine whether the question is generating the sort of information its author intends
 - Develop alternatives, if appropriate

To be addressed:

- What “additional verbal information” should we get?
- Who should we talk to and where should we find them?
- How do we make sense of the information we collect, in order to make better questions?

First: ask the question and observe the response

- Sometimes, respondents spontaneously provide evidence of a problem:
 - Expressions of confusion: “What does that even mean?”
 - Answers that don’t fit our intended response options
 - Responses that are clearly incorrect or contradictory
 - Don’t know and refusals
 - To make sense of these problems, we need *additional verbal information*
- Sometimes they provide an answer that seems adequate. However:
 - They might have answered based on misunderstandings or faulty assumptions about the meaning of the question
 - They might have had trouble remembering information, difficulty finding an adequate response option, or have provided a wild guess
 - To know if any of this is true, we need *additional verbal information*

What “additional verbal information” should we obtain?

- ▶ Original idea, from cognitive psychology: encourage participants to think out loud while answering questions (protocol analysis)
- ▶ In theory, gives us a window into cognitive processes and insight into what issues respondents are considering
- ▶ “In general, would you say your health is excellent, very good, good, fair, or poor?”

Think-aloud/protocol analysis

▶ Advantages:

- ▶ Straightforward for the interviewer, who simply provides prompting “remember to think out loud” as needed
- ▶ No potential for interviewer biasing the conversation

▶ However...

- ▶ Ability to verbalize thoughts varies widely: some respondents produce much more information than others
- ▶ Is it a realistic simulation of the survey– or does the act of explaining your thoughts change your thought processes?
- ▶ Is it valid? Do we literally have access to our thought processes, such that we can report them in real time?
- ▶ Is it practical? Does it give us information we need to identify and resolve problems?

Follow-up probing

- Alternative idea: use follow-up probes to assess quality of responses:
 - Ask the survey question, get an answer
 - Then, ask direct follow up questions, of several possible flavors:
 - What did you think the question meant (specific terms or overall)
 - How did you respond (what were you thinking or considering)
 - Explain your response (a larger narrative, give us more details or context)

Follow-up probing (2)

- Advantages of probing:
 - More direct– we ask exactly what we want to know
 - Seems to avoid the problems of varied respondent ability, and of “think-alouds” altering the response process
- However:
 - The interviewer may be adding a lot of material that the respondent might not have thought of– maybe differs from the actual survey experience
 - Probing can be leading
 - Interview content can vary a lot, unless we completely script all probes (which can be clunky)
 - This type of interviewing requires more skill
 - Contrasting tasks: succinct survey response; elaborate probe response

Balancing advantages and downsides of both

Ideally, employ a hybrid such that the interviewer adds *as little content to the interview as possible, but as much as necessary*

- 1) *encourage thinking aloud* to the extent possible
- 2) rely on *general, non-directive probing* (how did you come up with your answer, what did you mean by that) to generate more information
- 3) ask *more specific probes* if needed to help us understand how people came up with their answers
 - anticipate what probes might be useful in advance
 - don't use them if we don't need them; augment them if necessary

General probes may be sufficient

- “In general, would you say your health is excellent, very good, good, fair, or poor?”
- How did you pick your answer to this question?
- Tell me more about what you meant by your answer.
- Tell me more about your health in general.
- *Mirroring*: “Good?”

If we need to ask more directive probes

- Probe about interpretations of specific aspects of the question and particular respondent situations.
- “In general, would you say your health is excellent, very good, good, fair, or poor?”
 - “Have you had any illnesses or injuries lately?”
 - “When I say ‘excellent health’ what does that sound like it means to you?”
 - “What would it take for you to say that you had <one category higher> or <one category lower> health?”

Choose probes by considering various aspects of the question:

- Terms and concepts: any that are potentially unknown or ambiguous?
- Overall complexity: how long and complicated, how hard to understand overall?
- Recall task: how difficult will it be to remember what we're asking?
- Judgment: will people be willing and able to answer this?
- Response options: does the question provide reasonable categories/response format for answers?

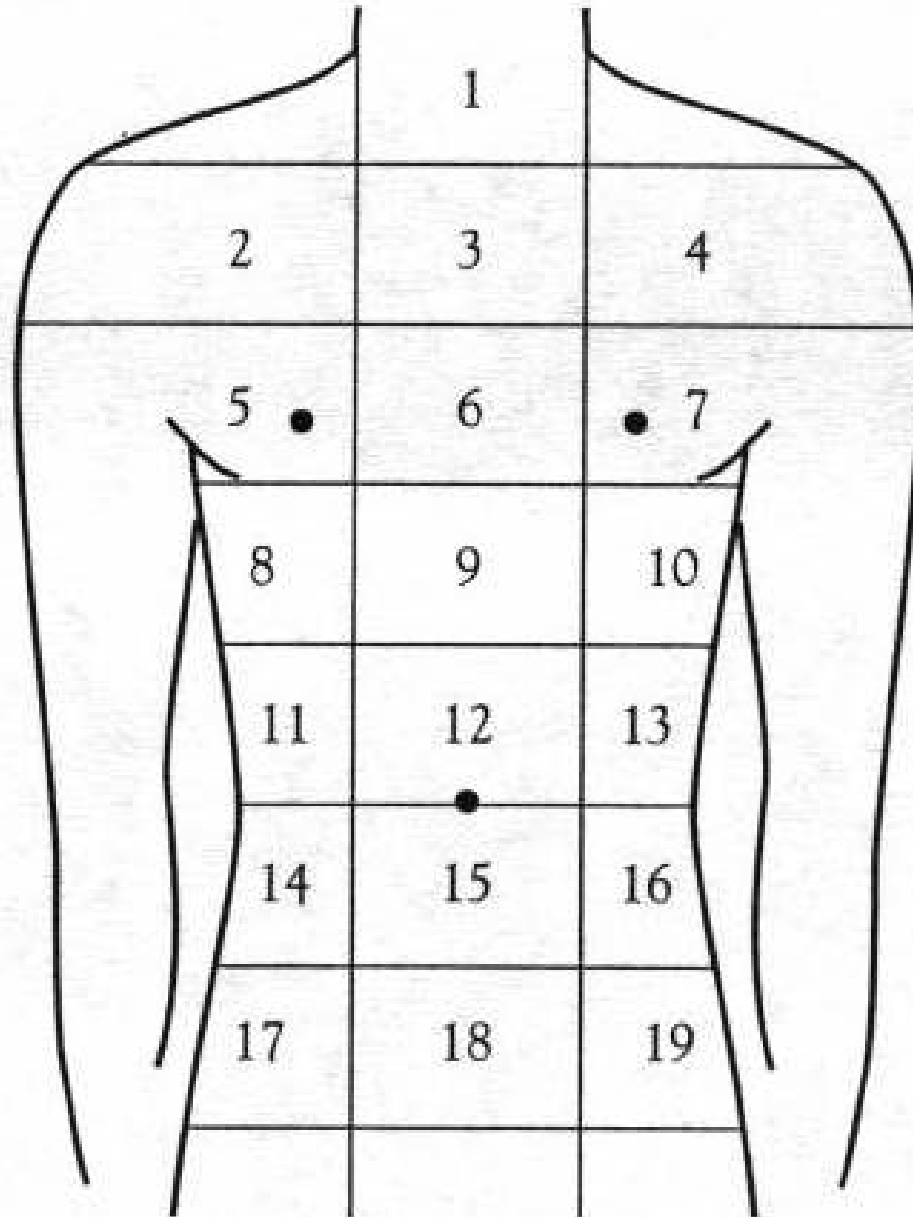
“In the last year, have you been bothered by pain in the abdomen?”

- Terms: is “abdomen” consistently understood?
- Judgment issues: how do they determine whether or not particular circumstance of pain was worth reporting?
- Memory issues: are such instances memorable and can their timing be remembered accurately?
- Possible probes that could address these concerns
 - Can you tell me what you think of as your abdomen?
 - Can you tell me about any pain or discomfort you’ve experienced around your stomach or the middle of your body?
 - What period of time are you thinking about here specifically?

“In the last year, have you been bothered by pain in the abdomen?”

- Encourage thinking out loud while answering. Afterwards, probe as necessary:
 - Can you tell me what you think of as your abdomen?
 - *In addition to narrative response, ask:* which of the numbered areas on this picture do you think of as the abdomen?
 - *Regardless of their earlier answer, ask:* Can you tell me about any pain or discomfort you’ve experienced around your stomach or the middle of your body?
- What period of time were you thinking about when answering?

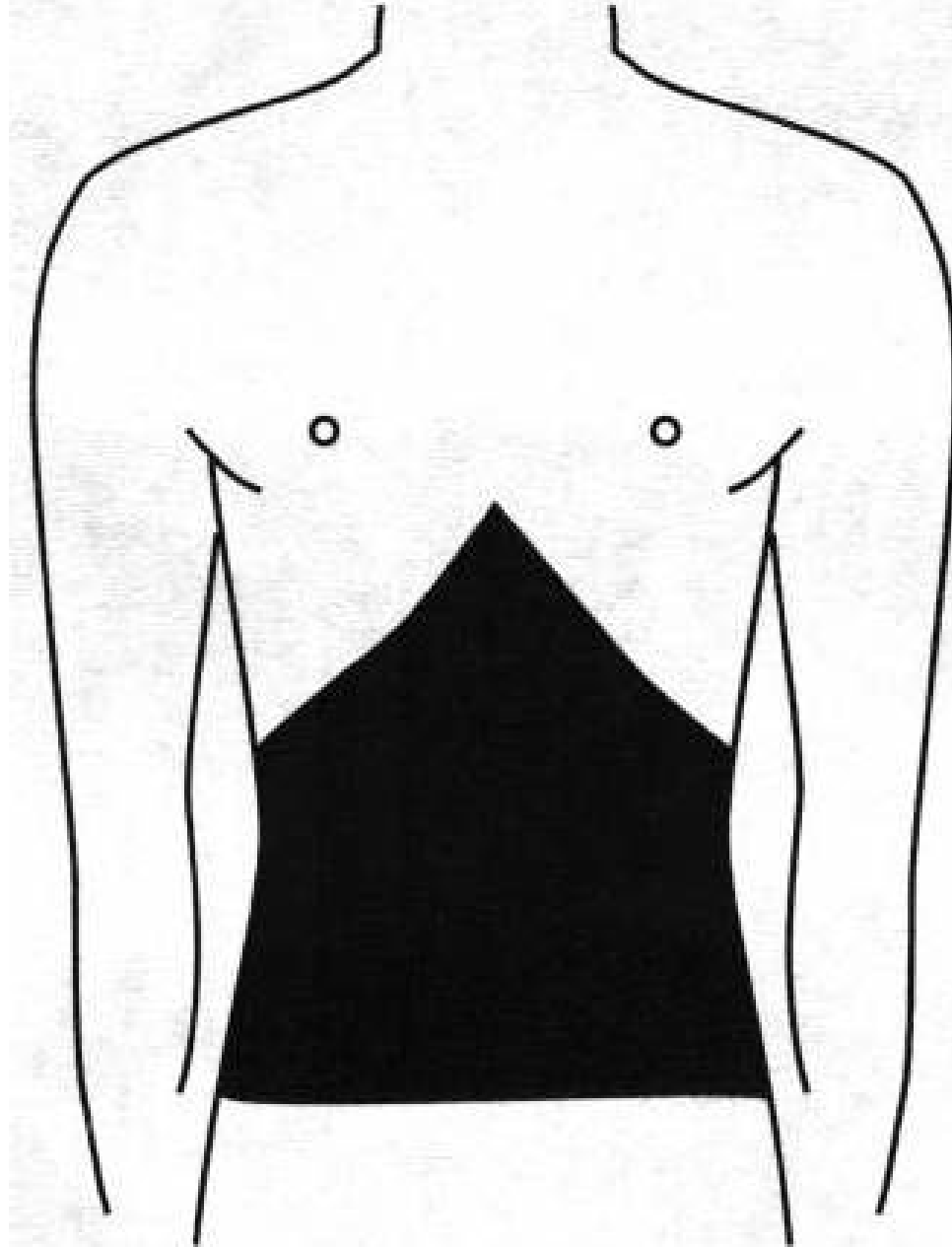
Can you tell me what you think of as your abdomen?



“In the last year, have you been bothered by pain in the abdomen?”

- Possible revisions:
 - Show shaded picture of abdomen
 - Drop “bothered”
 - Use “In the past 12 months”
- “In the past 12 months, have you had pain the abdomen? By abdomen, we mean the shaded area in this picture.”
- Our follow-up questions helped us to identify alternatives that
 - reduce ambiguity
 - lead to answers more consistent with our intent
 - with no apparent drawbacks

Shaded picture of the abdomen



When/how often should probing be done?

- Concurrent probes: between questions
 - Pros: Question is very fresh on the mind
 - Cons: Potential bias; switching of tasks can be distracting
- Retrospective probes: probe only at end of questionnaire
 - Pros: Avoids bias and task-switching
 - Cons: Long gap between question and probe

Two different visions of the interviewer role

- Investigator
 - has some understanding of questionnaire design and response errors
 - empowered to make decisions about probing
 - may provide great insight into issues difficult to predict in advance
 - may produce data hard to analyze systematically
 - potentially expensive
- Data collection agent
 - not necessarily any understanding of questionnaire design and response errors
 - primary task is to follow instructions and scripts determined by investigator
 - easier to find and train
 - produce more standardized data, though it may lack depth
 - design and analysis done by someone else

Summary: “additional verbal information”

- ▶ Before interviewing: evaluate the questionnaire, think about issues we might be concerned about, draft potential probes
- ▶ When interviewing, probe as unobtrusively as possible, but as much as needed (general -> specific)
- ▶ Use suggested probes as needed, and augment as necessary
- ▶ In earliest/most exploratory phases, probe concurrently (maximizing insight) ... later, probe less frequently as debriefing (maximizing realism)
- ▶ Use interviewers as knowledgeable and skilled as we can afford; methodological knowledge is a plus.

Mode of cognitive interview

- ▶ Cognitive interviews designed to be intensive and face-to-face; most survey interviews used to be that way
- ▶ Now self-administration is prevalent, and many surveys are multi-mode
- ▶ Face to face interviews may not be possible anyway
- ▶ What are best practices?

Mode of cognitive interview (2)

- ▶ Face to face cognitive interviewing is easier:
 - ▶ Communication
 - ▶ Rapport-building
 - ▶ Observation
- ▶ But it is extremely important for cognitive interview mode to mimic the intended survey mode
 - ▶ In-person cognitive interviewing might underestimate the challenges of asking questions by telephone
 - ▶ Paper or full-screen surveys might significantly alter the response process from a mobile device

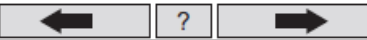
1. The following is a list of activities that people often do during the week. Although for some people with several medical problems it is difficult to determine what it is that limits them, please go over the activities listed below and indicate how much limitation you have had due to chest pain, chest tightness, or angina over the past 4 weeks:

Place an X in one box on each line

Activity	Extremely Limited	Quite a bit Limited	Moderately Limited	Slightly Limited	Not at all Limited	Limited for other reasons or did not do the activity
Dressing yourself	☐	☐	☐	☐	☐	☐
Walking indoors on level ground	☐	☐	☐	☐	☐	☐
Showering	☐	☐	☐	☐	☐	☐
Climbing a hill or flight of stairs without stopping	☐	☐	☐	☐	☐	☐
Gardening, vacuuming, or carrying groceries	☐	☐	☐	☐	☐	☐
Walking more than a block at a brisk pace	☐	☐	☐	☐	☐	☐
Running or jogging	☐	☐	☐	☐	☐	☐
Lifting or moving heavy objects (e.g. furniture, children)	☐	☐	☐	☐	☐	☐
Participating in strenuous sports (e.g. swimming, tennis)	☐	☐	☐	☐	☐	☐

The following is a list of activities that people often do during the week. Although for some people with several medical problems it is difficult to determine what it is that limits them, please go over the activities listed below and indicate how much limitation you have had **due to chest pain, chest tightness, or angina** over the past 4 weeks.

Tap -->.



Over the past 4 weeks:

Showering

Severely Limited

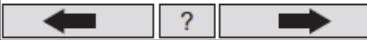
Moderately Limited

Somewhat Limited

A Little Limited

Not Limited

Limited, or did not
do for other reasons



Over the past 4 weeks:

Lifting or moving heavy objects
(e.g. furniture, children)

Severely Limited

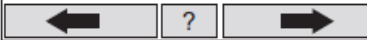
Moderately Limited

Somewhat Limited

A Little Limited

Not Limited

Limited, or did not
do for other reasons



Strategies for testing in different modes

- ▶ Laboratory-based telephone interviewing
- ▶ Laboratory-based self-administration (paper or web)
 - ▶ Preserves the ability to observe and record
 - ▶ Maintains realism
- ▶ Wireframe/mockup designs for mobile surveys
- ▶ Remote interviewing (Microsoft Teams, Zoom)
- ▶ Optimally: early interviews focus on communication, later ones focus on realism

Cognitive Interviewing Samples

- ▶ Often studies are conducted in rounds of about 12-15 cognitive interviews
 - ▶ After each round, take stock of problems identified, make adjustments to questionnaire
 - ▶ Conduct additional rounds with revised questionnaire until we are satisfied that problems are resolved— often 2 or 3 rounds in practice
- ▶ Larger sample needed when:
 - ▶ Questionnaires have different distinct branches that need to be tested
 - ▶ Different demographic groups might have meaningfully different experiences that need to be captured
- ▶ Samples not representative of the population in a statistical sense, but selecting respondents *appropriate to the survey* is very important

Recruitment considerations

- Appropriate to questionnaire content
- Demographic variety
- Not too skilled/experienced/educated relative to target population
- Don't "give away the questions" in advance during recruitment
- Allow time for challenges in finding people, plus any approval processes (IRB)
- Payment, especially if participants come to you

Where to find participants

- Advertisements (often online)
 - What the task entails (answering questions about a particular topic)
 - What characteristics they need to have
 - Logistics: when, where, how much time
 - Payment or other incentive
- Flyers, posted where relevant respondents are likely to be
- Community contacts
- Email lists
- Special interest organizations

Geographic diversity

- Traditionally, cognitive interviewing has been geographically limited due to practical constraints— generally done in locality of the researcher
- Over time we've realized this is inadequate for several reasons
 - Missing regional differences in experience and language
 - May be difficult to find special/rare populations
 - Raises numerous issues of diversity, equity and inclusion
- Travel as one solution, but expensive and time-consuming
- Pandemic taught us that remote testing is doable; technology (Zoom) has made it more feasible than before, in spite of some challenges
- Likely model going forward is a combination of local (in-person) and remote

What is an adequate sample size?

- Traditionally, samples have been limited by time and budgets (e.g., common to do several rounds of 12 interviews)
- Ideally, we would continue to interview until we stop learning new things
- Realistically, time and budgets usually run out first
- We assume that the most significant issues with a question are exposed in a relatively small number of interviews– hopefully reasonable if the sample is well-chosen
- Still, there is evidence that much larger sample sizes continue to yield important insights

One mitigation: web probing

- Simple approach to obtain large quantities of probe responses on web surveys
- Following a key survey question, interrupt with an open-ended probe (e.g. “Please tell us what sorts of things you included in your answer.”)
- Inexpensive and produces a high volume of data; no geographic limitation
- Limited to self-administered surveys
- Limited depth and quality of responses
- Use of web probing to augment traditional cognitive interviews is increasingly common

Follow up thoughts on sample sizes

- If we can't do as many interviews as needed, would we better off not doing any at all?
- Doing too few is better than doing none: the most significant problems are likely to be revealed in a small number of interviews
- Significance of a problem isn't wholly dependent upon how often it appears. More importantly: can we attribute the problem to some characteristic of the questionnaire?
- Our ideal is "saturation," but every interview toward that goal is likely to add value

Equipment

- Essentials: a questionnaire and a private space
- Highly desirable:
 - Means to administer the questionnaire in the intended mode
 - Audio recording (usually digital) with a good microphone
- Sometimes useful:
 - Video recording
 - Transcription
 - Processing/analytic software

Analyzing the Results

- Most analysis done at question level. Consider:
 - what ranges of answers did we see in this round
 - basis for each response– what was considered/remembered
 - evidence of disconnect from question intention based on the verbal exchanges through transcripts, recordings, notes
 - can we identify problematic aspects of questions
 - is there a viable alternative to be re-tested?
- Mostly qualitative: what is the evidence of a potential problem, more than how often it was observed
- Analysis is time consuming (possibly 3:1 writeup/analysis: interview)

Documenting results

- Sponsors may want to focus on the “bottom line”— what changes do you recommend to the questionnaire
- But it is also critical to document:
 - Interviewing protocol: what did you actually ask respondents to do
 - Recruitment: how did you find the people who were interviewed
 - Per question:
 - What range of answers did respondents provide?
 - What did people take into account while answering?
 - What problems were observed, if any, could threaten the validity of responses? What is your evidence that this is the case?
 - What alternatives might get around those problems?

In summary:

Administering cognitive interviews

1. Initially administer the question as it will be administered on the real survey (exact wording, preferably in actual mode of administration).
2. Encourage the respondent to answer the question in the expected format, if at all possible.
3. Use spontaneously offered information (think-aloud, hesitation, signs of confusion, clear errors) in initial evaluation.
4. When probing, start with broad narrative probes that explain responses...as necessary, move to more direct probes about interpretations/thought processes.

Administering cognitive interviews (2)

5. Use pre-scripted probes if relevant, but add to them or delete them as necessary and appropriate.
6. Talk less— encourage respondents to talk.
7. Avoid probes that suggest a particular answer/interpretation.
8. Consider probing by groups of questions rather than between individual questions, especially after initial interviews, to keep the flow realistic.

Administering cognitive interviewing (3)

9. If possible, use multiple interviewers to get different perspectives on how the questions are working. Ask each interviewer to contribute an independent analysis.
10. Record the interviews to aid analysis.
11. Analyze qualitatively. Study processes and the nature of responses rather than quantifying problems.
12. When you're convinced you've found a problem, try to fix it and test the revision.
13. Document your results.

Design and analysis: some examples

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Some research questions:

- ▶ What sorts of factors do people take into account when answering? Is it consistent with what we think of as “physical health”
- ▶ Is it easy to distinguish between days of “good” or “not good” health?
- ▶ How easy is it to remember health during the last 30 days?
- ▶ How easy is it to report a particular number of days?

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Data collection strategy:

- ▶ Administer the question, observe any response difficulties, encourage thinking out loud while answering
- ▶ If respondents don't provide a codeable response, ask again if they can respond with a number of days
- ▶ Probe as needed to supplement think-alouds and address research questions:
 - ▶ How did you decide on that number of days?
 - ▶ Please describe for me any illnesses or injuries that you included in your answer.
 - ▶ Did you have any difficulty deciding whether days were “good” or “not good”?
 - ▶ (Additional probing as needed to address particular responses).

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Recruitment:

- ▶ Seeking a mix of ages, but definitely wanted some “very young” (age 20-24) and “old” (age 75+); others in between
- ▶ Preferred respondents moderate to low SES
- ▶ Staff of a local senior center agreed to approach some residents to participate
- ▶ Initial round: 18 participants (9 male, 9 female)
- ▶ Median education 12th grade, but some college graduates, and one as low as 4th grade education

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Analysis:

- ▶ Interviewers independently observed that very few respondents initially provided direct numerical answers:
 - ▶ “I’ve been pretty good physically.”
 - ▶ “I’ve had a problem of the knee that’s been bothering me now for about four months.”
- ▶ Some could respond in days when prompted further, but others (particularly older respondents) expressed problems with the format:
 - ▶ “I can’t put a number of days on it.”
 - ▶ “It has nothing to do with days.”
- ▶ Most could describe physical health in terms of “all of the time,” “most,” “about half,” “none” etc.

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Analysis (2):

- ▶ It was helpful to categorize responses as follows:
 - ▶ Zero days, or “never”– no complaints; responding was straightforward
 - ▶ Thirty days, or “always”– felt bad all the time; responding straightforward
 - ▶ Specific numbers, e.g. “3 days”– discrete illnesses (cold) or flare up of a health condition; generally straightforward
 - ▶ Non-numeric response: usually complex health conditions that did not correspond to days
 - ▶ Some eventually provided a codeable response with considerable probing, but a few never did or explicitly rejected the premise of the question
- ▶ Respondents reported a variety of factors consistent with “physical health” as we understand it, and could easily keep track of the 30 day reference period, with appropriate examples

“Thinking about your own health, in last 30 days, for how many days would you say your physical health was not good?”

Conclusions:

- ▶ Respondents generally understood the meaning of the question
- ▶ Respondents found it easy to recall health status and health events in the 30-day reference period
- ▶ The response format proved challenging
 - ▶ Even in straightforward situations, initial responses were not numeric and required interviewer interpretation/intervention (e.g., “my health has been fine”)
 - ▶ Complex situations were difficult to report in days
 - ▶ Older respondents and those with complex conditions sometimes rejected the premise of “days” and never answered
- ▶ Are there alternatives that don’t use “days” as the metric?

At the present time, do you smoke cigarettes...

- 1) Daily
- 2) Occasionally,
- 3) Or not at all?

Main issue: do respondents understand the categories and find one that meets their situation?

At the present time, do you smoke cigarettes daily, occasionally, or not at all?

- A round of approximately 15 cognitive interviews included a wide range of smoking behavior
 - never smokers
 - former smokers but current nonsmokers
 - every day smokers
 - intermittent smokers
- All answered the question, none expressed confusion
- “It’s getting to be very occasionally...”
- Occasionally does not necessarily mean “less than daily”
- Alternative: “every day, less than every day, not at all.”

“Which of the following were you doing last week:”

- 1) working for pay at a job or business
 - 2) with a job or business but not at work
 - 3) looking for work
 - 4) working but not for pay at a job or business
 - 5) not working at a job or business and not looking for work
- Issues:
 - confusion about overall intent (activity last week vs employment status)
 - picking a response confusing for self-employed
 - not mutually exclusive (I had a job, but wasn't there last week because I was looking for another job)

Tradeoffs

In the past year, how many times have you talked with your doctor?

In the past year, how many times have you talked to a primary care or family physician about your own physical or mental health? Include telephone calls, but do not include visits for administrative purposes such as billing, or lab work.

Question wording to capture rare events

- In the past 12 months, how many times have you seen or talked on the telephone about your physical or mental health with a family doctor or general practitioner?
 - Common result: responses of “none/zero”
 - ... but contradicted by later responses in which they discussed recent visits to the doctor
 - This is a good example of an unanticipated problem calling for unscripted probing
 - Common reason: respondents misheard the question, thought it asked specifically about telephone contact only; question is too wordy
 - One reason questions are very complex is that their authors want to prompt respondents to think of a broadly inclusive range of situations, but this demonstrates a potential downside
 - Recommendation: say “seen or talked with” (drop “on the telephone”)

Experimental comparison

- “The next question is specifically about primary care doctors....”
 - V1: In the past 12 months, how many times have you seen or talked on the telephone with a primary care doctor about your health?
 - V2: In the past 12 months, how many times have you seen or talked with a primary care doctor about your health?
- The only difference between these two questions is the inclusion of “on the telephone.”

Results– Contact with doctors

	V1 (telephone)	V2 (no phone)	signif
Mean contacts	3.4	3.6	n.s.
“Zero” responses	24.7% (n=194)	9.2% (n=195)	p<.01
Resp requested help	5.7% (n=120)	11.3% (n=121)	p<.1

Maximizing the usefulness of the methodology

Common criticisms as cognitive interviewing has evolved

- Impressionistic
- Non-representative
- Misses many issues (at least with typical sample sizes); false negatives
- Uninformative about prevalence of issues
- Uninformative about severity of issues— difficult to distinguish trivial and deal-breaking errors
- Hasn't produced cumulative lessons learned

Problems and solutions

- Version 1: “In the last week, since [day], how many times did you eat vegetables, including corn, green beans, spinach, or any others?”
 - Evaluation: The examples tend to limit the vegetables they think of; recommend deleting to allow general frame of reference
- Version 2: “In the last week, since [day], how many times did you eat vegetables?”
 - Evaluation: Question fails to jog memory– addition of some examples would be helpful.

Making previous cognitive interview findings available

- One of the key obstacles to accumulation of findings has been difficulty finding past reports
- Increasingly, past tests are documented and available on line
- Databases of questionnaire evaluation reports such as Q-Bank: can search by question and access entire reports

Q-Bank

Improving surveys through sharing knowledge.

 Search

Reports

About Q-Bank

Archives

QUEST 2013 Workshop

Question Evaluation

Question Search

Search Q-Bank questions based on the question topic, survey title, testing agency, and/or keywords in the question text. Only questions that meet all of the selected criteria will be displayed in the search results.

Use the dropdown boxes and keywords to limit your search.

Question Search

Select **population** for questions about individuals or **establishment** for questions about organizations.

☐ Population

☐ Establishment

Question Topic

Survey Title

Testing Agency

Keyword

[Search](#)
[Help](#)

Search  Q-Bank

Use of previous findings

- ▶ Databases such as Q-Bank only include what has been submitted, and aren't necessarily "good" questions to use off the shelf
- ▶ However, it can provide full history of testing of certain items
- ▶ Finding past results doesn't necessarily eliminate the need for further testing...but it can save time and avoid "reinventing the wheel."
- ▶ wwwn.cdc.gov/qbank/home.aspx

The nature of question problems

Some researchers tend to treat problems in binary terms: either something is “a problem” that needs to be addressed, or not a problem

But almost always, questions have some degree of imperfection, some more serious than others. Deciding which issues warrant changes requires weighing tradeoffs.

Evaluating magnitude of imperfections

- How much does a supposed flaw matter in statistical terms, i.e., do distributions, variances, correlations change?
- More sophisticated experimentation needed to understand these consequences-- effortful and expensive to accomplish in practice
- On a more common basis, two criteria to evaluate magnitude of imperfections:
 - How widespread is the complication likely to be– how many will be affected by it?
 - When the issue comes up, how large of an impact is it likely to have on findings?

Example: evaluating magnitude of a problem

- “...for how many days of the last 30 days would you say your physical health was not good?”
 - Easy to answer for most:
 - if you are healthy with discrete exceptions
 - if always unhealthy (serious illness)
 - Difficult to answer for those with high variability within days, or constant low-level issues (common situation among elderly)
 - Zone of ambiguity: relatively small... question can be readily answered by most people
 - Impact: can be quite large if interested in elderly and chronic conditions... which the sponsors were

Making convincing cases to decision-makers

- 1) What is the nature of the problem?
- 2) What is the evidence that it exists?
- 3) What are the likely consequences of this problem (in terms of data quality) and how serious are they likely to be?
- 4) What viable alternatives are there?
- 5) (If known): What are the potential consequences of the proposed fix?

Final thoughts

- It is difficult for first-draft questions to adequately convey meaning
- Questionnaire design is a slow process, best performed in iterations, probably involving expert review, cognitive interview, field pretests, and other methods.
- The premise behind cognitive interviewing is straightforward: explore what respondents think while answering, through a combination of observation and additional questions.
- It tends to suggest design issues worthy of additional consideration– rarely definitive answers of the “best” way to ask a question.
- There are many methodological variations, but all can be helpful– the biggest barrier is usually allowing enough time.